Assignment #4

Due Date: Nov 24, 2025

1. *Elliptic Curve Cryptography*
   Elliptic Curve $E: y^2 \equiv x^3 + x + 7 \bmod 29$ is given.
   (a) Find any point α on the curve.
   (b) Compute the coordinates of the points $2\alpha, 3\alpha, 4\alpha$ and 5α on the given curve.
   (c) Find the coordinates of the points 8α by computing $3\alpha + 5\alpha$ and $4\alpha + 4\alpha$. Verify that you find the same point.
   (d) Find the number of points $\|E\|$ of the given curve.

2. *Elliptic Curve D-H Key Exchange*
   Alice and Bob want to share a key using D-H Key Exchange on Elliptic Curves. And, they choose the elliptic curve $E: y^2 = x^3 - x + 188 \bmod 751$ and a generator point $\alpha = (0, 376)$.
   Bob chooses $a_B = 5$ as the private key and Alice chooses $a_A = 3$.
   (a) Find the public keys for Alice and Bob.
   (b) Using D-H Key Exchange, find the common key generated by Alice and Bob.